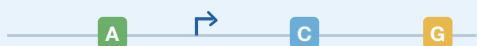


Individual 1

49kb around TSS



Individual 2

49kb around TSS



Pre-trained Enformer backbone

Convolutional tower

Transformer

Multi-scale feature extraction

At SNV positions across conv layers

SNVs and TSS

Query (Q): concatenate across individuals and multi-resolution bins

Sequence embeddings

Key (K), Value (V): concatenate across individuals

Contrastive cross-attention

With ROPE positional encoding



eQTL guidance

Auxiliary L_1 loss

Attention pooling

Aggregate variant and TSS effects



(Tissue-specific) MLP

Output heads



Blood

Δ Expression



Adipose

Δ Expression



Muscle

Δ Expression



CRZAM
PREDICTING PERSONALIZED GENE EXPRESSION
USING DNA SEQUENCE